

Jenkins Integration - Microservices

Step1 : Build the docker -compose file running the docker-compose.yml file.

```
master@master-vm:~/Microservices$ docker-compose up -d --build
WARN[0000] /home/master/Microservices/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it
to avoid potential confusion
Compose can now delegate builds to bake for better performance.
To do so, set COMPOSE_BAKE=true.
[+] Building 2.6s (31/31) FINISHED                                docker:default
=> [user-service internal] load build definition from Dockerfile    0.0s
=> => transferring dockerfile: 131B                                0.0s
=> [product-service internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 365B                                0.0s
=> [user-service internal] load metadata for docker.io/library/openjdk:2 1.8s
=> [user-service auth] library/openjdk:pull token for registry-1.docker. 0.0s
=> [payment-service internal] load metadata for docker.io/library/openjd 2.2s
=> [payment-service internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 291B                                0.0s
=> [order-service internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 365B                                0.0s
=> [user-service internal] load .dockerignore                      0.0s
=> => transferring context: 2B                                     0.0s
=> [user-service 1/3] FROM docker.io/library/openjdk:21@sha256:af9de795d 0.0s
=> [user-service internal] load build context                      0.0s
=> => transferring context: 86B                                    0.0s
=> CACHED [user-service 2/3] WORKDIR /app                          0.0s
=> CACHED [user-service 3/3] COPY target/*.jar app.jar             0.0s
=> [user-service] exporting to image                               0.0s
=> => exporting layers                                             0.0s
=> => writing image sha256:0db9ffd0a3f91422b91c3d68409f2dffb29ae8bd8870 0.0s
=> => naming to docker.io/bhargavakulla/user-service              0.0s
=> [user-service] resolving provenance for metadata file           0.0s
=> [order-service internal] load .dockerignore                    0.1s
=> => transferring context: 2B                                     0.0s
=> [product-service internal] load .dockerignore                  0.0s
=> => transferring context: 2B                                     0.0s
=> [payment-service internal] load .dockerignore                  0.1s
```

Step 2: Start the Services for each and microservices like order, payment, product and user-service and also for container and compressing the all source files and dependencies to maven.

```
✓ order-service          Built                                0.0s
✓ payment-service        Built                                0.0s
✓ product-service        Built                                0.0s
✓ user-service           Built                                0.0s
✓ Container order-service Started                        8.4s
✓ Container postgres_db  Started                        2.5s
✓ Container zookeeper    Started                        2.5s
✓ Container kafka_broker Started                        3.0s
✓ Container product-service Started                    4.3s
✓ Container user-service Started                        4.5s
✓ Container payment-service Started                    3.1s
master@master-vm:~/Microservices$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
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Step 3: Checking the Containers using docker ps command.

```
master@master-vm:~/Microservices$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
8922510bc9f4	bhargavakulla/order-service	"java -jar app.jar"	16 seconds ago	Up 6 seconds	8082/tcp, 0.0.0.0:8083->8083/tcp, :::8083->8083/tcp
983584f47388	bhargavakulla/payment-service	"java -jar app.jar"	16 seconds ago	Up 5 seconds	0.0.0.0:8084->8084/tcp, :::8084->8084/tcp
d8fbad325f4a	bhargavakulla/user-service	"java -jar app.jar"	16 seconds ago	Up 11 seconds	0.0.0.0:8081->8081/tcp, :::8081->8081/tcp
839b8af4375c	bhargavakulla/product-service	"java -jar app.jar"	16 seconds ago	Up 11 seconds	0.0.0.0:8082->8082/tcp, :::8082->8082/tcp
9bea0fac4348	confluentinc/cp-kafka:7.3.2	"/etc/confluent/dock..."	16 seconds ago	Up 10 seconds	0.0.0.0:9092->9092/tcp, :::9092->9092/tcp
3442be5a05a4	confluentinc/cp-zookeeper:7.3.2	"/etc/confluent/dock..."	17 seconds ago	Up 14 seconds	2888/tcp, 0.0.0.0:2181->2181/tcp, :::2181->2181/tcp
28e41855a572	postgres:17.4	"docker-entrypoint.s..."	17 seconds ago	Up 14 seconds	0.0.0.0:5432->5432/tcp, :::5432->5432/tcp

Step 4: Getting all the Kubernetes services pods.

```
master@master-vm:~/Microservices/k8s$ kubectl get all
```

NAME	READY	STATUS	RESTARTS	AGE
pod/flask-app-694d959f47-65blr	1/1	Running	1 (7m59s ago)	47h
pod/flask-app-694d959f47-7sqrg	1/1	Running	1 (7m59s ago)	47h
pod/flask-app-694d959f47-bvk8f	1/1	Terminating	0	6d2h
pod/flask-app-694d959f47-fs2c6	1/1	Terminating	0	6d2h
pod/flask-app-694d959f47-rlccq	1/1	Running	2 (7m59s ago)	6d2h
pod/flask-app-694d959f47-rwdt5	1/1	Running	2 (7m59s ago)	6d2h
pod/mysql-66d468f74c-6ggzz	1/1	Terminating	0	6d2h
pod/mysql-66d468f74c-bws29	1/1	Running	1 (7m59s ago)	47h
pod/nginx-78b5b78b99-7nwgk	1/1	Terminating	0	6d1h
pod/nginx-78b5b78b99-shswf	1/1	Running	1 (7m59s ago)	47h
pod/order-service-646676ccbf-m9cx9	0/1	CrashLoopBackOff	16 (2m23s ago)	44h
pod/order-service-7f79fc8645-h8sht	0/1	ImagePullBackOff	0	21s
pod/payment-service-5bf96544b8-l5v2m	0/1	CrashLoopBackOff	15 (4m48s ago)	44h
pod/payment-service-7674b6cc6c-tk52c	0/1	ContainerCreating	0	21s
pod/postgres-656894ccb5-pxz5v	0/1	ContainerCreating	0	20s
pod/product-service-6f9459ff7-4svvr	0/1	CrashLoopBackOff	15 (2m37s ago)	44h
pod/product-service-7665499cbc-f297c	0/1	ContainerCreating	0	20s
pod/user-service-67fd4d9c47-5p4q9	0/1	CrashLoopBackOff	13 (3m59s ago)	44h
pod/user-service-6bbc46cdb-hrrjd	0/1	ContainerCreating	0	20s

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/flask-service	ClusterIP	10.107.176.49	<none>	5000/TCP	6d2h
service/kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	6d2h
service/mysql	ClusterIP	10.110.104.236	<none>	3306/TCP	6d2h
service/nginx-service	NodePort	10.104.43.56	<none>	80:30007/TCP	6d1h
service/order-service	LoadBalancer	10.109.39.214	<pending>	8082:32549/TCP	44h
service/payment-service	LoadBalancer	10.110.136.173	<pending>	8084:30485/TCP	44h
service/postgres	ClusterIP	10.98.83.5	<none>	5432/TCP	21s
service/product-service	ClusterIP	10.107.79.210	<none>	8082/TCP	44h
service/user-service	LoadBalancer	10.99.120.93	<pending>	8081:30339/TCP	44h

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/flask-service	ClusterIP	10.107.176.49	<none>	5000/TCP	6d2h
service/kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	6d2h
service/mysql	ClusterIP	10.110.104.236	<none>	3306/TCP	6d2h
service/nginx-service	NodePort	10.104.43.56	<none>	80:30007/TCP	6d1h
service/order-service	LoadBalancer	10.109.39.214	<pending>	8082:32549/TCP	44h
service/payment-service	LoadBalancer	10.110.136.173	<pending>	8084:30485/TCP	44h
service/postgres	ClusterIP	10.98.83.5	<none>	5432/TCP	21s
service/product-service	ClusterIP	10.107.79.210	<none>	8082/TCP	44h
service/user-service	LoadBalancer	10.99.120.93	<pending>	8081:30339/TCP	44h

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
deployment.apps/flask-app	4/4	4	4	6d2h
deployment.apps/mysql	1/1	1	1	6d2h
deployment.apps/nginx	1/1	1	1	6d1h
deployment.apps/order-service	0/1	1	0	44h
deployment.apps/payment-service	0/1	1	0	44h
deployment.apps/postgres	0/1	1	0	20s
deployment.apps/product-service	0/1	1	0	44h
deployment.apps/user-service	0/1	1	0	44h

NAME	DESIRED	CURRENT	READY	AGE
replicaset.apps/flask-app-694d959f47	4	4	4	6d2h
replicaset.apps/mysql-66d468f74c	1	1	1	6d2h
replicaset.apps/nginx-78b5b78b99	1	1	1	6d1h
replicaset.apps/order-service-646676ccbf	1	1	0	44h
replicaset.apps/order-service-7f79fc8645	1	1	0	21s
replicaset.apps/payment-service-5bf96544b8	1	1	0	44h
replicaset.apps/payment-service-7674b6cc6c	1	1	0	21s
replicaset.apps/postgres-656894ccb5	1	1	0	20s
replicaset.apps/product-service-6f9459ff7	1	1	0	44h
replicaset.apps/product-service-7665499cbc	1	1	0	20s
replicaset.apps/user-service-67fd4d9c47	1	1	0	44h
replicaset.apps/user-service-6bbc46cdb	1	1	0	20s